

Discharging the $R4$ Branch from a Finite No-Multibubble Theorem

Abstract

This note isolates the final logical step in the $R4$ branch. We prove, in standard PDE language, that the generic $R4$ residual class is empty once finite retained active multibubble terminal states are excluded. We also state the precise finite no-multibubble theorem and show how it may be reused from a terminal nonlinear profile-decoupling theorem. The proof uses terminal state-space exhaustion, inactive-stratum discharge, recurrent tail rigidity, atomic sequence- L^3 completeness, mildness inheritance, and the Albritton–Barker Liouville theorem for mild ancient Navier–Stokes solutions.

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1 Renormalized setting

We work with the backward self-similar renormalized Navier–Stokes equation

$$\partial_\tau U - \Delta U + \frac{1}{2}y \cdot \nabla U + \frac{1}{2}U + (U \cdot \nabla)U + \nabla \Pi = 0, \quad \nabla \cdot U = 0 \quad (1.1)$$

on $\mathbb{R}_y^3 \times \mathbb{R}_\tau$. The associated physical variables are

$$u(x, t) = (-t)^{-1/2} U\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \quad t < 0, \quad (1.2)$$

and

$$p(x, t) = (-t)^{-1} \Pi \left(\frac{x}{\sqrt{-t}}, -\log(-t) \right). \quad (1.3)$$

A direct computation gives that (u, p) solves the physical Navier–Stokes equations

$$\partial_t u - \Delta u + (u \cdot \nabla) u + \nabla p = 0, \quad \nabla \cdot u = 0. \quad (1.4)$$

For a parabolic cylinder

$$Q_r(z_0) := B_r(y_0) \times (\tau_0 - r^2, \tau_0), \quad z_0 = (y_0, \tau_0),$$

define the local Caffarelli–Kohn–Nirenberg quantity

$$\mathcal{E}_r(U, \Pi; z_0) := \iint_{Q_r(z_0)} \left(|U|^3 + |\Pi - c_{r,z_0}(\tau)|^{3/2} \right) dy d\tau, \quad (1.5)$$

where $c_{r,z_0}(\tau)$ is an admissible local pressure gauge, for instance the average of $\Pi(\cdot, \tau)$ over $B_r(y_0)$. We write $Q_r = Q_r(0, 0)$.

2 The $R4$ class and terminal decomposition

The class $R4$ is the residual Type I branch left after all previously identified branches have been removed. Its elements are bounded ancient Seregini-limit profiles satisfying the Type I L^∞ bound, local suitability, persistent compact CKN density, and none of the previously closed small, stationary L^3 , tight, fast-decay, structured, axisymmetric, controlled-swirl, R1, R2, or R3 alternatives.

Definition 2.1 (R4 profile). *An R4 profile is a smooth bounded ancient solution (V, P) of (1.1) such that:*

- (i) $\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M$;
- (ii) V is suitable on every compact cylinder;
- (iii) V carries persistent compact CKN density, i.e. along the terminal extraction under consideration there is a compact cylinder on which the local CKN quantity is bounded below by a fixed positive number;
- (iv) V is not in any of the previously closed branches.

We assume the following terminal decomposition input.

Assumption 2.2 (Terminal state-space exhaustion). *Every terminal extraction of a tail-normalized R4 profile belongs to exactly one of the following alternatives:*

core, scattering, exterior, radiative, rough, finite multibubble.

Furthermore, the scattering, exterior, radiative, and rough alternatives do not retain the compact active CKN density of an R4 profile.

Assumption 2.3 (Tail reduction and affine normalization). *The tail-hull analysis has been performed. Nonconstant recurrent tail cores are discharged by recurrent tail rigidity, and constant tails are removed by the affine renormalized Galilean symmetry. Thus any remaining R4 candidate may be represented as a tail-zero profile.*

3 Retained active profiles and finite multibubbles

Fix a small positive threshold $\varepsilon_* > 0$.

Definition 3.1 (retained active terminal profile). *A terminal profile (U, Π) is called retained active if*

$$\mathcal{E}_1(U, \Pi; 0) \geq \varepsilon_*. \quad (3.1)$$

Definition 3.2 (finite retained active multibubble). *A finite retained active multibubble terminal state is a finite family of terminal cameras*

$$g_{1,n}, \dots, g_{J,n}, \quad J \geq 2,$$

such that:

(i) *the cameras are mutually separated:*

$$g_{i,n}^{-1} g_{j,n} \rightarrow \infty \quad (i \neq j);$$

(ii) *after passing to a subsequence,*

$$g_{j,n} \cdot V \rightarrow U_j$$

locally smoothly for velocity and locally in $L^{3/2}$ for pressure modulo admissible gauges;

(iii) *each terminal profile is retained active:*

$$\mathcal{E}_1(U_j, \Pi_j; 0) \geq \varepsilon_*, \quad j = 1, \dots, J.$$

The packet is called maximal if no further separated retained active terminal camera may be added to it.

Theorem 3.3 (Finite no-multibubble theorem). *No R4 profile admits a finite retained active multibubble terminal state.*

Theorem 3.3 is the final multibubble statement. In Section 9, we explain how it can be reused from terminal nonlinear profile decoupling.

4 Atomic terminal profiles

Definition 4.1 (tail descendant). *Let (U, Π) be a terminal profile. A retained active tail descendant of U is a terminal limit W obtained from cameras escaping to infinity relative to U , such that*

$$\mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

Definition 4.2 (radiative and rough descendants). *A radiative descendant of U is a sequence*

$$\tau_n \rightarrow -\infty, \quad R_n \rightarrow \infty,$$

and a number $\eta > 0$, such that

$$\int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq \eta,$$

while every unit-scale exterior camera is below the retained-active threshold, after excluding rough cameras.

A rough descendant is a tail camera for which the local compactness, suitability, or pressure control needed to extract a terminal profile fails.

Definition 4.3 (atomic profile). *A retained active terminal profile is atomic if it has no retained active tail descendant, no radiative descendant, and no rough descendant.*

5 One active descendant gives a two-camera multibubble

Lemma 5.1. *Let U be a retained active terminal profile. If U has a retained active tail descendant, then U belongs to the finite retained active multibubble stratum.*

Proof. The origin camera of U carries positive local CKN mass:

$$\mathcal{E}_1(U, \Pi; 0) \geq \varepsilon_*.$$

Let W be a retained active tail descendant. By definition there are cameras $h_n \rightarrow \infty$, relative to U , such that

$$h_n \cdot U \rightarrow W$$

in the terminal topology and

$$\mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

By local convergence of the velocity and pressure, after gauges,

$$\mathcal{E}_1(h_n \cdot U, \Pi_{h_n}; 0) \geq \frac{1}{2} \varepsilon_*$$

for all sufficiently large n . Thus, for large n , the same terminal object has two separated retained active cameras: the origin camera and the camera h_n . This is a finite two-camera retained active multibubble. $\square$

Proposition 5.2. *Assume Theorem 3.3 and Assumption 2.2. Then every singleton retained active terminal profile arising from an RA candidate is atomic.*

Proof. Let U be a singleton retained active terminal profile. If U had a retained active tail descendant, Lemma 5.1 would place it in the finite multibubble stratum, contradicting Theorem 3.3.

If U had a radiative or rough descendant, then the corresponding terminal extraction would lie in the radiative or rough alternative. These alternatives are excluded as carriers of the persistent compact CKN density by Assumption 2.2. Hence no such descendant exists, and U is atomic. $\square$

6 Atomic sequence- L^3 completeness

Lemma 6.1. *Let U be a bounded smooth ancient solution of (1.1). If U is retained active and atomic, then there exists a sequence*

$$\tau_k \rightarrow -\infty$$

such that

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. We argue by contrapositive. Suppose no such sequence exists. Then

$$\liminf_{\tau \rightarrow -\infty} \|U(\cdot, \tau)\|_{L^3(\mathbb{R}^3)} = \infty. \tag{6.1}$$

Choose $\tau_n \rightarrow -\infty$ so that

$$\int_{\mathbb{R}^3} |U(y, \tau_n)|^3 dy \rightarrow \infty.$$

Since U is bounded, choose $R_n \rightarrow \infty$ sufficiently slowly so that

$$\int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq 1. \tag{6.2}$$

Set

$$m_n := \sup_{|x| > R_n} \int_{B_1(x)} |U(y, \tau_n)|^3 dy.$$

If $\limsup_n m_n \geq \varepsilon_*$, then for a subsequence there exist $|x_n| > R_n$ such that

$$\int_{B_1(x_n)} |U(y, \tau_n)|^3 dy \geq \varepsilon_*.$$

After recentering at (x_n, τ_n) , either terminal compactness fails, which gives a rough descendant, or a terminal subsequence converges to a retained active tail descendant. Both alternatives contradict atomicity.

It remains to consider the case $m_n < \varepsilon_*$ for all sufficiently large n . Then (6.2) says that exterior L^3 -mass persists, while every unit-scale exterior camera is below the retained-active threshold. This is precisely a radiative descendant, contradicting atomicity.

Thus (6.1) is impossible. $\square$

7 Albritton–Barker endpoint

Assumption 7.1 (mildness inheritance). *If U is a retained active terminal profile, then the physical pullback (1.2) is a mild ancient solution of the physical Navier–Stokes equations, after any finite time shift.*

Theorem 7.2 (Albritton–Barker). *Let u be a mild ancient solution of the three-dimensional Navier–Stokes equations. If there exists a sequence $t_k \downarrow -\infty$ such that*

$$\sup_k \|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then $u \equiv 0$.

Lemma 7.3. *Assume Assumption 7.1. No retained active atomic terminal profile exists.*

Proof. Let U be retained active and atomic. By Lemma 6.1, choose $\tau_k \rightarrow -\infty$ such that

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3} < \infty.$$

Let u be the physical pullback (1.2), and set

$$t_k = -e^{-\tau_k}.$$

Then $t_k \downarrow -\infty$, and by critical scaling,

$$\|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} = \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)}.$$

By Assumption 7.1, u is mild ancient. Theorem 7.2 therefore gives $u \equiv 0$, hence $U \equiv 0$. This contradicts retained activity, since $\mathcal{E}_1(U, \Pi; 0) \geq \varepsilon_* > 0$. $\square$

8 R4 closure from no multibubbles

Theorem 8.1 (R4 closure from finite no-multibubble). *Assume:*

- (i) *tail reduction and affine normalization, Assumption 2.3;*
- (ii) *terminal state-space exhaustion and inactive-stratum discharge, Assumption 2.2;*

- (iii) no finite retained active multibubbles, Theorem 3.3;
- (iv) mildness inheritance, Assumption 7.1;
- (v) the Albritton–Barker Liouville theorem, Theorem 7.2.

Then the $R4$ class is empty.

Proof. Suppose, for contradiction, that $V \in R4$. By Assumption 2.3, we reduce to a tail-zero representative. By definition of $R4$, persistent compact CKN density remains.

Apply terminal state-space exhaustion. By Assumption 2.2, the persistent compact CKN density cannot be carried by scattering, exterior, radiative, or rough alternatives. Hence it is carried either by a finite retained active multibubble terminal state or by at least one singleton retained active terminal profile.

The first case is impossible by Theorem 3.3. In the second case, let U be such a singleton retained active terminal profile. By Proposition 5.2, U is atomic. By Lemma 7.3, U cannot exist. This contradiction proves that no $R4$ profile exists. $\square$

9 Reusing a no-multibubble theorem

The proof above isolates exactly how a no-multibubble theorem is reused in the $R4$ branch. The input required by Theorem 8.1 is simply Theorem 3.3. One may prove Theorem 3.3 by any method. A standard route is through terminal nonlinear profile decoupling.

Theorem 9.1 (Terminal nonlinear profile decoupling). *Let (V_n, P_n) be a terminal sequence with a maximal finite retained active packet $g_{1,n}, \dots, g_{J,n}$. Then, after passing to a subsequence, there are mild ancient profiles (U_j, Π_j) , $j = 1, \dots, J$, such that:*

- (i) $g_{j,n} \cdot V_n \rightarrow U_j$ in C_{loc}^∞ ;
- (ii) $g_{j,n} \cdot P_n - c_{j,n}(\tau) \rightarrow \Pi_j$ locally in $L^{3/2}$;
- (iii) in the j -th camera, every other profile escapes every compact set;
- (iv) the pressure generated by distant profiles is harmonic on compact cylinders and is absorbed into the local pressure gauge;
- (v) nonlinear cross-interactions between distinct profiles vanish on compact terminal windows;
- (vi) after removing the finite packet, no hidden retained active camera remains.

Theorem 9.1 is the usual terminal no-splitting statement: separated active profiles behave independently in terminal cameras, and any further retained active camera contradicts maximality.

Proposition 9.2 (Terminal decoupling implies no finite multibubbles). *Assume Theorem 9.1, discharge of radiative and rough terminal alternatives, recurrent tail rigidity, mildness inheritance, and the Albritton–Barker theorem. Then Theorem 3.3 holds.*

Proof. Suppose a finite retained active multibubble packet exists, and let $U_1, \dots, U_J$ be a maximal packet.

If some U_j is atomic, then Lemma 7.3 gives a contradiction.

Assume therefore that no U_j is atomic. Since radiative and rough alternatives are discharged, every U_j has a retained active tail descendant. By Theorem 9.1, each such descendant lifts back to a retained active terminal camera of the original sequence. By maximality of the

packet, the lifted camera must be equivalent to one of the existing $g_{i,n}$. Thus the finite set of indices $\{1, \dots, J\}$ carries a directed graph in which each vertex has an outgoing edge.

A finite directed graph with every vertex having an outgoing edge contains a directed cycle. This cycle produces a compact recurrent tail set. By recurrent tail rigidity, that recurrent tail set consists only of affine constant profiles. After tail-zero normalization, the recurrent set is trivial. This contradicts retained activity of the profiles in the cycle.

Therefore the assumption that a finite retained active multibubble exists is impossible. $\square$

Combining Proposition 9.2 with Theorem 8.1 gives the final reusable form:

terminal nonlinear profile decoupling $\implies$ no finite retained multibubbles $\implies R4 = \emptyset$.
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10 Summary of the closure chain

The complete logical chain is:

$$\begin{aligned}
&\text{terminal nonlinear profile decoupling} \\
&\implies \text{no finite retained active multibubbles in } R4 \\
&\implies \text{all singleton retained active carriers are atomic} \\
&\implies \text{backward sequence-}L^3 \text{ completeness} \\
&\implies \text{Albritton–Barker Liouville contradiction} \\
&\implies R4 = \emptyset.
\end{aligned}$$

References

References

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